

String Amplitudes Project

Damià Canyameres Morante, Micheal Doyle and Paul Haskins

canyamed@tcd.ie, doylem55@tcd.ie, haskinsp@tcd.ie

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1 Changes

- Make "Background" follow the flow of ie prescription in qft, ie string (witten), ie Jan, Binachi. Not generic Polchinski/Tong.
- Plot and verify differing values from Bianchi
- Cleanup
- NB: Have a conclusions
- Mod forms?
- Keep old version as a high level conceptual overview

2 Background

As we know from QFT (relativistic 0+1 dimensional particles), scattering amplitudes are fundamental objects that encode the probabilities of various particle interactions. In the context of string theory (1 dimensional objects), scattering amplitudes are computed using the world sheet formalism, where the interactions of strings are described by two-dimensional conformal field theories. The dynamics of string interactions are governed by the Nambu-Goto action, or equivalently, the Polyakov action, both of which are given below:

$$\mathcal{S}_{NG}[X] = -\frac{1}{2\pi\alpha'} \int_{\mathcal{M}} d\tau d\sigma \sqrt{-\det(\partial_a X^\mu \partial_b X_\mu)} \quad (2.1)$$

$$\mathcal{S}_P[X, \gamma] = -\frac{1}{4\pi\alpha'} \int_{\mathcal{M}} d\tau d\sigma \sqrt{-\gamma} \gamma^{ab} \partial_a X^\mu \partial_b X_\mu \quad (2.2)$$

where $X^\mu(\tau, \sigma)$ are the embedding coordinates of the string world sheet $\mathcal{M}$ in the target spacetime, γ_{ab} is the world sheet metric, and α' is the Regge slope parameter related to the string tension. The Polyakov action is invariant under reparametrisations of the world sheet coordinates and local Weyl transformations, which allows us to fix a gauge and simplify calculations.

Varying with respect to X^μ gives us the equation of motion. Imposing Neumann boundary conditions for open strings and Dirichlet boundary conditions for closed strings, we can solve the equations of motion to obtain the mode expansions of the string coordinates. The quantization of these modes leads to the spectrum of string states, which include both massless and massive excitations. The massless states correspond to gauge bosons and gravitons, while the massive states correspond to higher-spin particles.

A natural question that arises is: how to compute scattering amplitudes for these string states? The computation of scattering amplitudes in string theory involves evaluating correlation functions of vertex operators on the world sheet. Vertex operators are constructed to represent the emission/absorption of string states and are locally inserted on the world sheet. The correlation functions of these vertex operators are computed using the path integral formalism, where we integrate over all possible world sheet configurations weighted by the exponential of the Polyakov action. The resulting scattering amplitudes are then expressed in terms of integrals over the moduli space of Riemann surfaces, which encode the different topologies of string interactions.

The S -matrix element is thus given by

$$S_{j_1 \dots j_n}(k_1, \dots, k_n) = \sum_{\substack{\text{compact} \\ \text{topologies}}} \int \frac{[dX dg]}{V_{\text{diff} \times \text{Weyl}}} e^{-S_X - \lambda \chi} \prod_{i=1}^n \int d^2 \sigma_i g(\sigma_i)^{\frac{1}{2}} \mathcal{V}_{j_i}(k_i, \sigma_i) \quad (2.3)$$

where χ is the Euler characteristic of the world sheet, λ is the string coupling constant, and $\mathcal{V}_{j_i}(k_i, \sigma_i)$ are the vertex operators corresponding to the external string states with momenta k_i and polarizations j_i . The sum over compact topologies accounts for all possible world sheet configurations that contribute to the scattering process.

2.1 World Sheet Configurations (Compact Topologies)

One runs into issues with consistency when trying to sum over all world sheet configurations bounded by arbitrary initial/final curves. Local world sheet symmetries cause overcounting of physically equivalent configurations, which leads to divergences in the path integral.

To solve this issue we postulate that we can take the limit where string sources are at infinity. This corresponds to an S matrix element where we have specified the incoming/outgoing strings.

At infinity we have free propagation of strings, which allows us to identify both the incoming and outgoing legs with a cylinder given by

$$-2\pi t \leq \Im w \leq 0, \quad w \sim w + 2\pi \quad (2.4)$$

In general, the sum over topologies is not restricted to connected world sheets. However, it is more convenient to consider them. In two dimensions, the classification of compact connected oriented surfaces with boundaries are topologically equivalent to a sphere with g handles and b holes.

For unoriented surfaces, define the crosscap as a disk with antipodal points identified. Any compact connected closed surface, (oriented or not) can be obtained by adding g handles and c crosscaps to a sphere. The Euler characteristic of a surface with g handles, b boundaries and c crosscaps is given by

$$\chi = 2 - 2g - b - c \quad (2.5)$$

2.2 Vertex Operators

In QFT, there exists the space of states $\mathbf{S}$ and the space of local operators $\mathbf{O}$. Conformal symmetry allows us to identify an isomorphism $\mathbf{S} \cong \mathbf{O}$ where the CFT is quantised on a circle. If we consider the semi-infinite cylinder $\mathbb{R} \times S^1$ with coordinates $w = \tau + i\sigma$, we can map the cylinder to the complex plane via the conformal transformation $z = e^{iw}$. The point at infinity on the cylinder corresponds to the origin on the complex plane, and the point at negative infinity on the cylinder corresponds to the point at infinity on the complex plane.

To define the path integral in the w coordinate, we need to specify boundary conditions at $\tau = \infty$ and $\tau = -\infty$. The boundary conditions at $\tau = \infty$ corresponds to the insertion of a local operator at the origin of the complex plane *known as the Vertex Operator associated with that state*, while the boundary conditions at $\tau = -\infty$ corresponds to the insertion of a local operator at infinity on the complex plane. Conversely, the path integral on the disk with an operator $\mathcal{A}$ at the origin maps to the path integral on the cylinder with a specified state $|\mathcal{A}\rangle$.

In the context of the path integral description, the fields ϕ have a fixed value at the boundary of the disk ϕ_b . The path integral over the fields ϕ_i in the interior of the disk with ϕ_b fixed gives a functional,

$$\Psi(\phi_b) = \int [d\phi_i]_{\phi_b} e^{-S[\phi_i]} \mathcal{A}(0) \quad (2.6)$$

. Conversely, given a state $\Psi[\phi_b]$. One can consider a path integral over the annulus $1 \geq |z| \geq r$, with the fields ϕ_b on the outer circle fixed and the field ϕ'_b integrated over the inner circle

$$\int [d\phi_i]_{\phi_b, \phi'_b} e^{-S[\phi_i]} r^{L_0 + \bar{L}_0} \Psi[\phi'_b] \quad (2.7)$$

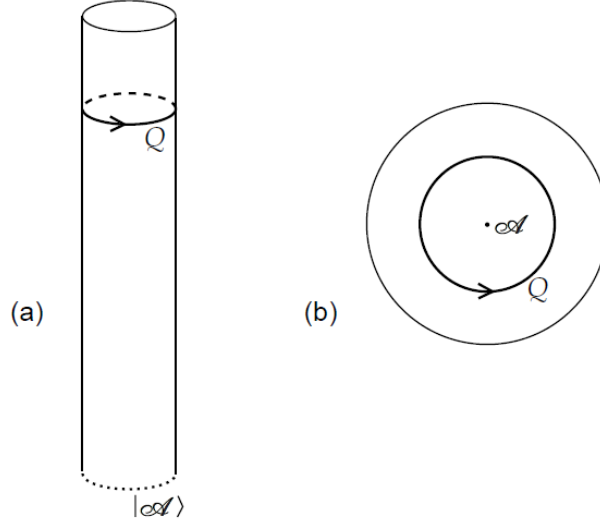


Fig. 2.6. (a) A semi-infinite cylinder in w , with initial state $|\mathcal{A}\rangle$ and charge Q . (b) The conformally equivalent unit disk with local operator $\mathcal{A}$ and contour integral for Q .

Figure 2.1: The semi-infinite cylinder maps under $z = e^{iw}$ to the punctured disc, with the point at $\tau = \infty$ sent to the origin. An operator inserted at the origin of the disc is equivalent to a fixed boundary state on the cylinder.

where L_0 and $\tilde{L}_0$ are the Virasoro generators. The integral is weighted by the functional $r^{L_0 + \tilde{L}_0}$. The integral over the annulus corresponds to propagating from $|z| = r$ to $|z| = 1$ which is equivalent to acting by our functional. This reverses the operator acting on Ψ , so 2.7 gives back $\Psi[\phi_b]$. We note that the path integral over the annulus is equivalent to a path integral over the cylinder, which is equivalent to a path integral over the disk with an operator inserted at the origin. This establishes the isomorphism between states and operators in CFT.¹ One can also consider the Polyakov formalism where we define the path integral on the Polyakov action

$$Z = \int \frac{\mathcal{D}X \mathcal{D}g}{V_{\text{diff} \times \text{Weyl}}} e^{-\mathcal{S}_P[X, g]}. \quad (2.8)$$

If we consider a perturbation $\mathcal{S}_P \rightarrow \mathcal{S}_P + \lambda \int d^2z \mathcal{O}(z, \bar{z})$. The $\mathcal{O}$ is a possible local operator.²

2.3 Weyl Invariance and Conformal Transformations

In going from the Nambu-Goto action 2.1 to the Polyakov action 2.2 we obtained a new symmetry, Weyl invariance (2d)

$$X'^{\mu}(\tau, \sigma) = X^{\mu}(\tau, \sigma) \quad \gamma'_{ab}(\tau, \sigma) = e^{2\omega(\tau, \sigma)} \gamma_{ab}(\tau, \sigma) \quad (2.9)$$

¹Excited states may be obtained by acting on an operator with the derivative operator.

²To represent a physical state it must obey Weyl invariance.

where ω is arbitrary.

This determines γ_{ab} up to a local rescaling, so Weyl invariant metrics map to the same embedding in spacetime. Thus in the context of the path integral we need to quotient by Weyl transformations.

Likewise for new coordinates (τ', σ') we have a diffeomorphism invariance

$$X'^{\mu}(\tau, \sigma) = X^{\mu}(\tau, \sigma) \quad \frac{\partial \sigma'^c}{\partial \sigma^a} \frac{\partial \sigma'^d}{\partial \sigma^b} \gamma'_{ab}(\tau', \sigma') = \gamma_{ab}(\tau, \sigma) \quad (2.10)$$

which we will also need to quotient by in the path integral.

2.4 Equivalent Representations of Incoming and Outgoing States

The first viewpoint assumes we are given boundary conditions at the ends of the world sheet where we prescribe initial and final string shapes and the world sheet represents unitary evolution between them.

Another interpretation is that at infinity strings propagate freely and the world sheet is a cylinder

$$-2\pi t \leq \Im w \leq 0, \quad w \sim w + 2\pi \quad (2.11)$$

$w \sim w + 2\pi$ means the spatial direction around the string is periodic, so the external leg is cylindrical.

Finally, after a conformal transformation, we can map the cylinder to a punctured disk, where the puncture corresponds to the point at infinity. A vertex operator inserted at a puncture represents the emission or absorption of a string state, and the correlation functions of these vertex operators give us the scattering amplitudes.

In §3 we will replicate the results of [1], in which, they prove the equivalence of the $i\epsilon$ prescription and regularisation via generalised exponentials. Then, in §4 we replicate the results of [2], in which, they implement said equivalence to calculate the one loop mass corrections to super-string massive higher-spin states.

3 Explicit Replication of $i\epsilon$ Method for String Amplitudes

Here we will replicate the results and intermediate steps from the paper [1].

Decomposing $\mathcal{F}_Y$ into $\mathcal{F}_1$ and a compact rectangle

$$R_Y = \left\{ x \in \left[-\frac{1}{2}, \frac{1}{2} \right], y \in [1, Y] \right\} \quad (3.1)$$

Namely, $q = e^{2\pi i\tau}$, $\bar{q} = e^{-2\pi i\bar{\tau}}$, $\tau \in \mathbb{H}$. $\tau := x + iy$, $\iff x = \frac{\tau + \bar{\tau}}{2}$, $y = \frac{\tau - \bar{\tau}}{2i}$

$$\begin{aligned} L_{m,n,s}(Y) &= \int_{\mathcal{F}_1} d\tau \wedge d\bar{\tau} y^{-s} q^m \bar{q}^n - 2i \int_{\frac{1}{2}}^{-\frac{1}{2}} \int_1^Y dx \wedge dy y^{-s} q^m \bar{q}^n \\ L_{m,n,s}(Y) &= \int_{\mathcal{F}_1} d\tau \wedge d\bar{\tau} y^{-s} q^m \bar{q}^n - 2i\delta_{m,n} (E_s(4\pi m) - Y^{1-s} E_s(4\pi m Y)) \end{aligned}$$

Proof:

Taking the second term,

$$-2i \int_{\frac{1}{2}}^{-\frac{1}{2}} \int_1^Y dx \wedge dy y^{-s} q^m \bar{q}^n = -2i \int_{-\frac{1}{2}}^{\frac{1}{2}} dx e^{2\pi i(m-n)x} \int_1^Y dy y^{-s} e^{-2\pi(m+n)y}.$$

Consider the integral over x for $m \neq n$

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} dx e^{2\pi i(m-n)x} = \frac{1}{2\pi i(m-n)} \left[e^{\pi i(m-n)} - e^{-\pi i(m-n)} \right] = 0.$$

for $m = n$ it is 1. Therefore, we have

$$-2i \int_{\frac{1}{2}}^{-\frac{1}{2}} \int_1^Y dx \wedge dy y^{-s} q^m \bar{q}^n = -2i\delta_{m,n} \int_1^Y dy y^{-s} e^{-2\pi(m+n)y} \quad (3.2)$$

$$-2i\delta_{m,n} \int_1^Y dy y^{-s} e^{-2\pi(m+n)y} = -2i\delta_{m,n} \int_1^Y dy y^{-s} e^{-4\pi(m)y} \quad (3.3)$$

Let $t = 4\pi m y$.

$$\begin{aligned} & -2i\delta_{m,n} \int_{4\pi m}^{4\pi m Y} t^{-s} e^{-t} dt \\ & -2i\delta_{m,n} \left[(4\pi m)^{s-1} \int_{4\pi m}^{\infty} e^{-t} t^{-s} dt - Y^{1-s} (4\pi m Y)^{s-1} \int_{4\pi m Y}^{\infty} e^{-t} t^{-s} dt \right] \\ & = -2i\delta_{m,n} (E_s(4\pi m) - Y^{1-s} E_s(4\pi m Y)). \end{aligned}$$

3.1 The Equivalence

To demonstrate the equivalence of the amplitudes $\mathcal{A}_0^{i\epsilon}$ and $\mathcal{A}_0^r$ they claim that

$$L_{m,n,s}^{i\epsilon} = L_{m,n,s}^r \quad (3.4)$$

where

$$\begin{aligned} L_{m,n,s}^{i\epsilon} &= \int_{\mathcal{F}_0} d\tau \wedge d\bar{\tau} y^{-s} q^m \bar{q}^n + \int_{T_0}^{T_0+i\infty} dy y^{-s} q^m \bar{q}^n \\ L_{m,n,s}^r &= \int_{\mathcal{F}_0} d\tau \wedge d\bar{\tau} y^{-s} q^m \bar{q}^n - 2i\delta_{m,n} T_0^{1-s} E_s(4\pi m T_0). \end{aligned}$$

Thus we need to show

$$\int_{T_0}^{T_0+i\infty} dy y^{-s} e^{-4\pi m y} = T_0^{1-s} E_s(4\pi m T_0) \quad (3.5)$$

where the rhs is given by

$$T_0^{1-s} E_s(4\pi m T_0) = (4\pi m)^{s-1} \int_{4\pi m T_0}^{\infty} dt t^{-s} e^{-t} \quad (3.6)$$

If we consider the case where $m < 0$ and $s \geq 1$. To avoid the multivaluedness of the t^{-s} factor, we take the branch cut along the negative real axis and infinitesimally deform the contour into the lower half plane.

Define

$$J_l(m, R) = \int_{T_0}^{T_0+iR} dy y^{-s} e^{-4\pi m y} \quad (3.7)$$

$$J_r(m, R) = \int_{T_0}^{T_0+\text{sgn}(m)R} dy y^{-s} e^{-4\pi m y}, \quad \text{sgn}(m) = \begin{cases} = 1 & \text{if } m \geq 0 \\ = -1 & \text{if } m < 0 \end{cases}. \quad (3.8)$$

Showing that, in the limit $R \rightarrow \infty$, (3.7) equals (3.8) is equivalent to verifying that 3.5 holds. To prove said restatement, we define the contour integral around a quadrant of the circle centered at $y = T_0$.

$$J(m, R) = J_l(m, R) - J_r(m, R) + J_\phi(m, R) \quad (3.9)$$

where we use $y(\phi) := T_0 + R e^{i\phi}$ to define the *arc integral* J_ϕ

$$J_\phi(m, R) = \begin{cases} \int_{\frac{\pi}{2}}^{\pi} dy(\phi) y(\phi)^{-s} e^{-4\pi m y(\phi)} & \text{if } m < 0 \\ \int_{\frac{\pi}{2}}^0 dy(\phi) y(\phi)^{-s} e^{-4\pi m y(\phi)} & \text{otherwise} \end{cases} \quad (3.10)$$

We note that since $J(m, R)$ vanishes, if J_ϕ vanishes in the $\lim_{R \rightarrow \infty}$ then we have $\lim_{R \rightarrow \infty} J_l(m, R) = \lim_{R \rightarrow \infty} J_r(m, R)$ as required.

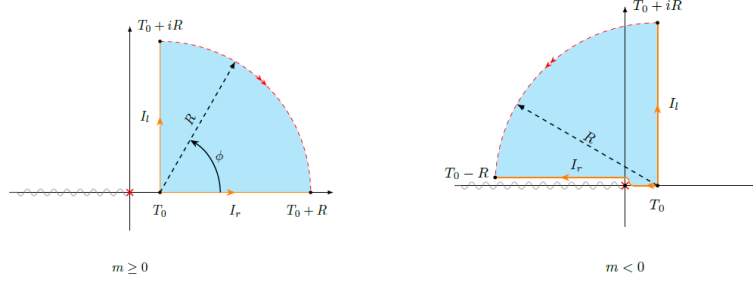


Figure 3.1: Contour of integration which avoids any singularities and thus $J(m, R)$ must vanish by Cauchy's theorem [1].

Thus, consider the case of $m < 0$,

$$\begin{aligned}
 |J_\phi(m, R)| &= \left| \int_{\frac{\pi}{2}}^{\pi} dy(\phi) y(\phi)^{-s} e^{-4\pi m y(\phi)} \right| \\
 |J_\phi| &= e^{-4\pi m T_0} R^{1-s} \left| \int_{\frac{\pi}{2}}^{\pi} d\phi \left(\frac{T_0}{R} + e^{i\phi} \right)^{-s} e^{4\pi m R e^{i\phi}} \right| \\
 |J_\phi| &= e^{-4\pi m T_0} R^{1-s} \left| \int_{\frac{\pi}{2}}^{\pi} d\phi \left(\frac{T_0}{R} + e^{i\phi} \right)^{-s} e^{4\pi m R (\cos \phi + i \sin \phi)} \right| \\
 |J_\phi| &\leq e^{-4\pi m T_0} R^{1-s} \int_{\frac{\pi}{2}}^{\pi} d\phi \left| \frac{T_0}{R} + e^{i\phi} \right|^{-s} e^{4\pi m R \cos \phi}
 \end{aligned}$$

Within our contour ($\phi \in [\frac{\pi}{2}, \frac{3\pi}{2}]$), one has $e^{-4\pi m R \cos \phi} \leq 1$

$$|J_\phi| \leq e^{-4\pi m T_0} R^{1-s} \int_{\frac{\pi}{2}}^{\pi} d\phi \left| \frac{T_0}{R} + e^{i\phi} \right|^{-s}.$$

Thus, for $s > 1$, the limit as $R \rightarrow \infty$ of J_ϕ vanishes and thus we have shown that $\lim_{R \rightarrow \infty} J_l(m, R) = \lim_{R \rightarrow \infty} J_r(m, R)$ as required.¹ This same technique also holds for the simpler case of $m \geq 1$ where there are no tachyonic divergence issues.

[Open question not yet resolved] The bound above only covers $s > 1$; the case $s \leq 1$, where the arc estimate $|J_\phi| \rightarrow 0$ no longer obviously holds, still needs to be worked out.

¹Note m is bounded below by $m = -1$

4 Replicating the One-Loop Mass Corrections of Bianchi et al.

4.1 Integration Over the Worldsheet Coordinate

The level matching condition requires

$$N_1 + N_2 = 2(N - 1) = \bar{N}_1 + \bar{N}_2. \quad (4.1)$$

$$\begin{aligned} \mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \int_0^1 dx \int_0^1 dy e^{-4\pi(N-1)\tau_2 y^2} \theta_1(x + \tau y | \tau)^{N_1} \overline{\theta_1(x + \tau y | \tau)}^{\bar{N}_1} \\ &\quad \theta_2(x + \tau y | \tau)^{N_2} \overline{\theta_2(x + \tau y | \tau)}^{\bar{N}_2}. \end{aligned}$$

Recalling the relation $k^2 = \sum_i k_i^2$, and $k = (k_1, \dots, k_{2(N-1)})$, and using eqn (4.16) of [2], we have

$$\begin{aligned} \mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \int_0^1 dx \int_0^1 dy e^{-4\pi(N-1)\tau_2 y^2} \sum_{k, \bar{k} \in (\mathbb{Z} + \frac{1}{2})^{2(N-1)}} q^{\frac{1}{2}k^2} e^{2\pi i \sum_i k_i(x + \tau y)} (-1)^{\sum_i (k_i - \frac{1}{2})a_i} \\ &\quad \bar{q}^{\frac{1}{2}\bar{k}^2} e^{-2\pi i \sum_i \bar{k}_i(x + \bar{\tau} y)} (-1)^{\sum_i (\bar{k}_i - \frac{1}{2})\bar{a}_i}. \end{aligned}$$

With the constraints

$$\sum_{i=1}^{2(N-1)} k_i = \sum_{i=1}^{2(N-1)} \bar{k}_i = \sum_{i=1}^{2(N-1)} \bar{k}_i = H \in \mathbb{Z}, \quad (4.2)$$

where $H = h + 2(N-1)M$ with $h \in \{0, \dots, 2N-3\}$ and $M \in \mathbb{Z}$. If we use the notation of Bianchi

$$k_i = l + k_i \quad \bar{k}_i = l + \bar{k}_i, \quad (4.3)$$

so,

$$\begin{aligned} \mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \sum_{\substack{\sum_i k_i = \sum_i \bar{k}_i = H}} \int_0^1 dy e^{-4\pi(N-1)\tau_2 y^2} q^{\frac{1}{2}k^2} e^{-4\pi\tau_2 y} (-1)^{\sum_i (k_i - \frac{1}{2})a_i} \\ &\quad \bar{q}^{\frac{1}{2}\bar{k}^2} e^{-2\pi i \bar{\tau} y H} (-1)^{\sum_i (\bar{k}_i - \frac{1}{2})\bar{a}_i} \\ &= \sum_M \sum_{h=0}^{2N-3} \sum_{\substack{\sum_{i=1}^{2(N-1)} k_i = \sum_{i=1}^{2(N-1)} \bar{k}_i \\ = h + 2M(N-1)}} \int_0^1 dy e^{-4\pi(N-1)\tau_2 (y+M)^2 + 4\pi(N-1)\tau_2 M^2 - 4\pi\tau_2 y h} \\ &\quad q^{\frac{1}{2}k^2} \bar{q}^{\frac{1}{2}\bar{k}^2} (-1)^{\sum_i (k_i - \frac{1}{2})a_i + \sum_i (\bar{k}_i - \frac{1}{2})\bar{a}_i}. \end{aligned}$$

which gives,

$$\begin{aligned}
\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \sum_{\sum_i k_i = \sum_i \bar{k}_i = h} q^{\frac{1}{2} \sum_i k_i^2} \bar{q}^{\frac{1}{2} \sum_i \bar{k}_i^2} (-1)^{\sum_i (k_i - \frac{1}{2}) a_i + \sum_i (\bar{k}_i - \frac{1}{2}) \bar{a}_i} \int_{-\infty}^{\infty} dy e^{-4\pi(N-1)\tau_2 y^2 - 4\pi\tau_2 y h} \\
\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2} &= \sum_{\sum_i k_i = \sum_i \bar{k}_i = h} q^{\frac{1}{2} \sum_i k_i^2} \bar{q}^{\frac{1}{2} \sum_i \bar{k}_i^2} (-1)^{\sum_i (k_i - \frac{1}{2}) a_i + \sum_i (\bar{k}_i - \frac{1}{2}) \bar{a}_i} \underbrace{\int_{-\infty}^{\infty} dy e^{-4\pi(N-1)\tau_2 \left(y + \frac{h}{N-1}\right)^2}}_{\frac{1}{\sqrt{(N-1)\tau_2}}} e^{\frac{4\pi}{N-1}\tau_1 h^2} \\
&= \frac{1}{\sqrt{(N-1)\tau_2}} \Theta_{A_{2N-3h}} (-1)^{\sum_i (k_i - \frac{1}{2}) a_i + \sum_i (\bar{k}_i - \frac{1}{2}) \bar{a}_i}
\end{aligned}$$

EQN (4.16)

$$\begin{aligned}
\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2}(\tau, \bar{\tau}) &:= \frac{(-1)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{\tau_2}(N-1)} \sum_{h=1}^{2(N-1)} \sum_{\mathbf{K} \in \mathbb{Z}^{2(N-1)}} \sum_{\bar{\mathbf{K}} \in \mathbb{Z}^{2(N-1)}} \delta_{\mathbf{1}_{2(N-1)} \cdot \mathbf{K}, h} \delta_{\mathbf{1}_{2(N-1)} \cdot \bar{\mathbf{K}}, h} \\
&\quad q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4(N-1)}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \bar{q}^{\frac{1}{2} \bar{\mathbf{K}} \cdot \bar{\mathbf{K}} - \frac{h^2}{4(N-1)}} e^{-i\pi \bar{\mathbf{a}} \cdot \bar{\mathbf{K}}}
\end{aligned} \tag{4.4}$$

Taking the case $N_1 = \bar{N}_1 = 2$ and $N_2 = \bar{N}_2 = 0$, which satisfies level matching as required, and substituting in accordance with this choice, we obtain the following expression for the master integral.

$$\begin{aligned}
\mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2}(\tau, \bar{\tau}) &:= \frac{(-1)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{\tau_2}} \sum_{\mathbf{K} \in \mathbb{Z}^2} \sum_{\bar{\mathbf{K}} \in \mathbb{Z}^2} \delta_{\mathbf{1}_2 \cdot \mathbf{K}, h} \delta_{\mathbf{1}_2 \cdot \bar{\mathbf{K}}, h} \\
&\quad q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \bar{q}^{\frac{1}{2} \bar{\mathbf{K}} \cdot \bar{\mathbf{K}} - \frac{h^2}{4}} e^{-i\pi \bar{\mathbf{a}} \cdot \bar{\mathbf{K}}},
\end{aligned}$$

where the first summation has been dropped due to the particular choice of $N = 2$. We note that, although we have fixed the dimension of the lattice over which we are summing, infinitely many sites are still considered in the sum. This appears in the integral for the amplitude $\mathcal{A}^{(2)}$ for which we wish to compute,

$$\mathcal{A}^{(2)} = \int_{\mathcal{F}} \frac{d^2\tau}{y^2} \frac{\mathcal{I}_{2,2,0,0}(\tau, \bar{\tau})}{|\vartheta_1'(0|\tau)|^4}.$$

where the master integral is now

$$\begin{aligned}
\mathcal{I}_{2,2,0,0}(\tau, \bar{\tau}) &:= \frac{1}{2\sqrt{\tau_2}} \sum_{\mathbf{K} \in \mathbb{Z}^2} \sum_{\bar{\mathbf{K}} \in \mathbb{Z}^2} \delta_{\mathbf{1}_2 \cdot \mathbf{K}, h} \delta_{\mathbf{1}_2 \cdot \bar{\mathbf{K}}, h} \\
&\quad q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \bar{q}^{\frac{1}{2} \bar{\mathbf{K}} \cdot \bar{\mathbf{K}} - \frac{h^2}{4}} e^{-i\pi \bar{\mathbf{a}} \cdot \bar{\mathbf{K}}},
\end{aligned}$$

having substituted the desired values for the indices. We now impose the constraints given by the Kronecker deltas by defining

$$k \equiv K_1, \quad K_2 = h - k, \tag{4.5}$$

and similarly for the barred variables

$$\bar{k} \equiv \bar{K}_1, \quad \bar{K}_2 = h - \bar{k}. \quad (4.6)$$

Since the inner sums are over a 2-dimensional lattice and recalling the definitions of q and $\bar{q}$ from [1],

$$q = e^{2\pi i \tau} \text{ and } \bar{q} = e^{-2\pi i \bar{\tau}}, \quad (4.7)$$

where

$$\tau = \tau_1 + i\tau_2 \text{ and } \bar{\tau} = \tau_1 - i\tau_2, \quad (4.8)$$

we may simplify the exponents of the factors in the above expression. Firstly, let us compute the exponent of

$$q^{\frac{1}{2}\mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4}}. \quad (4.9)$$

It is, after completing the square, given by

$$\begin{aligned} \frac{1}{2}\mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4} &= 1/2(k^2 + (h-k)^2) - h^2/4 \\ &= 1/2(k^2 + k^2) + 1/2h^2 - 1/4h^2 - kh \\ &= k^2 + 1/4h^2 - kh \\ &= (k+h)^2 - 2kh, \end{aligned} \quad (4.10)$$

which, for convenience, we define as

$$\alpha_{k,h} \equiv (k+h)^2 - 2kh. \quad (4.11)$$

The calculation with the barred variables proceeds similarly to give

$$\beta_{\bar{k},h} \equiv (\bar{k}+h)^2 - 2\bar{k}h. \quad (4.12)$$

By the definition of q , we have

$$q^{\frac{1}{2}\mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4}} = e^{2\pi i \alpha_{k,h} \tau} = e^{2\pi i \alpha_{k,h} (\tau_1 + i\tau_2)} \quad (4.13)$$

and

$$\bar{q}^{\frac{1}{2}\bar{\mathbf{K}} \cdot \bar{\mathbf{K}} - \frac{\bar{h}^2}{4}} = e^{-2\pi i \beta_{\bar{k},h} \bar{\tau}} = e^{2\pi i \beta_{\bar{k},h} (-\tau_1 + i\tau_2)}. \quad (4.14)$$

The other factors can be pulled out of the resulting integral, as they do not depend on τ ,

$$e^{i\pi \mathbf{a} \cdot \mathbf{K}} = e^{i\pi(a_1 k + a_2(h-k))}, \quad e^{-i\pi \bar{\mathbf{a}} \cdot \bar{\mathbf{K}}} = e^{i\pi(\bar{a}_1 \bar{k} + \bar{a}_2(h-\bar{k}))}. \quad (4.15)$$

Assembling all of this gives the integral

$$A = \Sigma_k \Sigma_{\bar{k}} B_{k,h} \int d\tau_1 \int d\tau_2 \left(e^{-2\pi(\alpha_{k,h} + \beta_{\bar{k},h})\tau_2} \right) \frac{1}{|\theta'_1(0|\tau)|^4}, \quad (4.16)$$

where we use the following identity for the derivative of the Jacobi theta function,

$$\theta'_1(0|\tau) = 2\pi\eta(\tau)^3. \quad (4.17)$$

Since in the $N = 2$ case we are summing over the values $h = 1, 2$, we can relate some of the expressions above to Jacobi theta functions by inserting $1 = e^{2\pi i k}$ and comparing with the definition. By inspection, we have for $h = 1$ that

$$\begin{aligned} \sum_{k \in \mathbb{Z}} q^{(k+1/2)^2} e^{i\pi} &= \sum_{l \in \mathbb{Z}+1/2} q^{l^2} e^{2\pi i(l-1/2)} e^{i\pi} \\ &= \sum_{l \in \mathbb{Z}+1/2} q^{l^2} e^{2\pi i l} \\ &= \theta_2(2\tau, 1) \\ &= -\theta_2(2\tau, 0). \end{aligned} \quad (4.18)$$

Similarly, for the other iteration of the sum, $h = 2$, we have

$$\begin{aligned} \sum_{k \in \mathbb{Z}} q^{(k-1)^2} e^{2i\pi} &= \sum_{l \in \mathbb{Z}} q^{l^2} e^{2\pi i l} \\ &= \theta_3(\tau, 0). \end{aligned} \quad (4.19)$$

[Unverified needs checking] Combining the two cases above suggests

$$|\theta_3(2\tau)|^2 + |\theta_2(2\tau)|^2, \quad (4.20)$$

but the argument for this combination hasn't been checked carefully yet and the conditions under which it holds haven't been stated.

4.2 Root Lattice Structure in the Computation of the Master Integral

From the equation for the master integral, it can be shown that there is a root lattice structure from the infinite sums over the vectors $\mathbf{K}$ and $\bar{\mathbf{K}}$. In particular, we see an A_{2N-3} root lattice for different N . This is owing to the fact that we have $2(N-1)$ a priori degrees of freedom for each lattice (barred and unbarred), restricted by one constraint in each case, namely the Kronecker delta. To see this, let us work inductively starting with the $N = 2$ case. In particular, we wish to justify why we need only compute $I_{2,2,0,0}$ in order to obtain the scattering amplitude, which in principle is given by a sum over the values of $N_1, N_2, \bar{N}_1, \bar{N}_2$, restricted by the level matching condition $N_1 + N_2 = \bar{N}_1 + \bar{N}_2$.

For $N = 2$, the function $Q(k) = 2k^2$ appears in the sum over the lattice Γ ,

$$\sum_{\mathbf{K} \in \Gamma} q^{1/2 Q(\mathbf{K})} = \sum_{\mathbf{K} \in \Gamma} q^{k^2}. \quad (4.21)$$

We can make a table of the allowed values of $N_1, N_2, \bar{N}_1, \bar{N}_2$. In the right-most column, we display the factor by which each contributing term in the sum differs.

The factors in the right column are simply the contribution from

$$(-1)^{\frac{N_1 + \bar{N}_1}{2}} \quad (4.22)$$

Table 4.1: Possible values of $N_1, N_2, \bar{N}_1, \bar{N}_2$

N_1	$\bar{N}_1$	N_2	$\bar{N}_2$	Factor
2	2	0	0	1
2	1	0	1	-i
1	0	1	2	i
0	0	2	2	1
1	1	1	1	-1
1	2	1	0	-i
0	1	2	1	i

in (4.4). It can be seen that, other than this factor, there is no difference in the individual terms of the sum, since the exponential factors yield 1 for any $2\pi l$ in their arguments with l an integer. This allows us to take $N_1 = \bar{N}_1 = 2$, i.e. $I_{2,2,0,0}$, up to an overall factor. Adding the coefficients, we get -1 for this overall factor on $I_{2,2,0,0}$.

Next, we turn our attention to the $N = 3$ case. Here, the restriction imposed by the Kronecker delta is

$$k_1 + k_2 + k_3 + k_4 = h. \quad (4.23)$$

For $h = 0$, this allows us to simplify the notation by letting

$$k_4 = -(k_1 + k_2 + k_3). \quad (4.24)$$

The quadratic $Q(k)$ appearing in the exponential is now given by

$$\begin{aligned} \mathbf{K} \cdot \mathbf{K} &= (k_1)^2 + (k_2)^2 + (k_3)^2 + (-(k_1 + k_2 + k_3))^2 \\ &= 2((k_1)^2 + (k_2)^2 + (k_3)^2 + k_2 k_3 + k_1 k_3 + k_1 k_2). \end{aligned} \quad (4.25)$$

If we construct a matrix from the coefficients of this quadratic, we may note that its determinant is precisely the number of independent elements of the root lattice,

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 2(3) - 1(1) + 1(-1) = 4 = 2N - 2 \text{ for } N = 3, \quad (4.26)$$

that is, the determinant of the coefficients of the quadratic in the exponential from the master integral formula (4.4) matches the number of independent elements of the lattice before the Kronecker delta constraint is applied.

4.3 The Decay Width of the Renormalised Amplitude

We have the expression for the amplitude given by

$$\mathcal{A}_{\text{ren}}^{(N)} := \mathcal{A}_{\mathcal{F}_1}^{(N)} + \sum_{\ell=0}^{2(N-1)} \sum_{k_\ell=0}^{\nu_\ell} \frac{c_{\ell,k_\ell}^{(N)}}{\sqrt{N-1}} E_{\frac{9}{2}+\ell} \left(-\frac{\pi b_{k_\ell}^{(N)}}{N-1} \right) \quad (4.27)$$

where ν_ℓ is the number of identical negative powers of q and $\bar{q}$, $c_{\ell,k_\ell}^{(N)}$ is the coefficient relating the weight of the various internal channels which contribute to the self energy state, $b_{k_\ell}^{(N)}$ is a positive integer depending on the identical negative powers of q and $\bar{q}$, and $\mathcal{A}_{\mathcal{F}_1}^{(N)}$ is the finite part contributing to the mass shift,

$$\mathcal{A}_{\mathcal{F}_1}^{(N)} := \int_{-\frac{1}{2}}^{\frac{1}{2}} d\tau_1 \int_{\sqrt{1-\tau_1^2}}^1 \frac{d\tau_2}{|\vartheta_1'(0|\tau)|^{4(N-1)}} \sum_{k=0}^{N-2} \binom{N-2}{k}^2 \sum_{n,\bar{n}=0}^{N-2} \binom{k}{n} \binom{k}{\bar{n}} \quad (4.28)$$

$$\tilde{e}_1(\tau)^n \overline{\tilde{e}_1(\tau)}^{\bar{n}} f_1(\tau)^{k-n} \overline{f_1(\tau)}^{k-\bar{n}} \mathcal{I}_{2n+2,2\bar{n}+2,2N-4,2N-4-2\bar{n}}(\tau, \bar{\tau}). \quad (4.29)$$

If we look at the second contribution to the decay width we consider

$$\Im A_{\mathcal{F}_T} := \Im \left[\sum_{\ell=0}^{2(N-1)} \sum_{k_\ell=0}^{\nu_\ell} \frac{c_{\ell,k_\ell}^{(N)}}{\sqrt{N-1}} E_{\frac{9}{2}+\ell} \left(-\frac{\pi b_{k_\ell}^{(N)}}{N-1} \right) \right] \quad (4.30)$$

where

$$E_s(z) := \begin{cases} z^{s-1} \int_z^\infty dt e^{-t} t^{-s} & \text{if } z \in \mathbb{C} \setminus \{0\} \\ \frac{1}{s-1} & \text{if } z = 0 \text{ and } s \neq 1 \\ 0 & \text{otherwise} \end{cases} \quad (4.31)$$

. In order to determine whether the generalised exponential is complex we can use the identity for when the argument is non-vanishing.

$$E_s(z) = z^{s-1} \Gamma(1-s, z) \quad (4.32)$$

Since the argument is real we only need to determine its sign. For $z \geq 0$, then clearly z^{s-1} , and $e^{-t} t^{-s}$ lie on the real line.

For $z < 0$ we can let $z := -x$ for some $x > 0$. Then we have

$$z^{s-1} = e^{(s-1)\ln(z)} \quad \ln(-x) = \ln(x) + i\pi \quad (4.33)$$

$$\begin{aligned} (-x)^{s-1} &= e^{(s-1)(\ln(x)+i\pi)} \\ &= x^{s-1} e^{i\pi(s-1)} \\ &= x^{s-1} [\cos(\pi(s-1)) + i \sin(\pi(s-1))]. \end{aligned}$$

We note that the integral $\Gamma(s-1, z)$ is real. We are given that b_{k_ℓ} is positive so the argument is negative $\forall N$. Thus we can write the net contribution of $E_{\frac{9}{2}+\ell} \left(-\frac{\pi b_{k_\ell}^{(N)}}{N-1} \right)$ as

$$E_{\frac{9}{2}+\ell} \left(-\frac{\pi b_{k_\ell}^{(N)}}{N-1} \right) = x^{\frac{7}{2}+\ell} \left[\cos\left(\pi\left(\frac{7}{2}+\ell\right)\right) + i \sin\left(\pi\left(\frac{7}{2}+\ell\right)\right) \right] \quad (4.34)$$

where the first term drops out since $\ell \in \mathbb{Z}$. Thus we can define $R_\ell := \left[\frac{\pi b_{k_\ell}^{(N)}}{N-1} \right]^{\frac{7}{2}+\ell} \sin \left(\pi \left(\frac{7}{2} + \ell \right) \right)$ and write

$$E_{\frac{9}{2}+\ell} \left(-\frac{\pi b_{k_\ell}^{(N)}}{N-1} \right) = i R_\ell \quad (4.35)$$

We can identify the form of $b_{k_\ell}^{(N)}$ by comparing with (eq 5.5)

$$4\pi m = -\frac{\pi b_{k_\ell}^{(N)}}{N-1}, \quad \implies b_{k_\ell}^{(N)} = 4m(1-N) \text{ and } m = -\frac{b_{k_\ell}^{(N)}}{4(N-1)} \quad (4.36)$$

4.4 Implementing the $i\epsilon$ Prescription

Bianchi eqn [2] (4.18)

$$\mathcal{A}^{(N)} = \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^{\frac{9}{2}} |\vartheta'_1(0|\tau)|^{4(N-1)}} \sum_{k=0}^{N-2} \binom{N-2}{k}^2 \sum_{n, \bar{n}=0}^{N-2} \binom{k}{n} \binom{k}{\bar{n}} \tilde{e}_1(\tau)^n \overline{\tilde{e}_1(\tau)}^{\bar{n}} f_1(\tau)^{k-n} \overline{f_1(\tau)}^{k-\bar{n}} \mathcal{I}_{2n+2, 2\bar{n}+2, 2N-4, 2N-4-2\bar{n}}(\tau, \bar{\tau}).$$

4.4.1 The Weierstrass Contribution $W(q, \bar{q})$

We define $\tilde{e}_1(\tau) := e_1(\tau) - \mathcal{E}(\tau)$ and e_{s-1} are the Weierstrass invariants and $\mathcal{E}(\tau)$ is a quasi-holomorphic modular form

$$e_{s-1}(\tau) = -4\pi i \partial_\tau \log \frac{4\vartheta_s(0, \tau)}{\eta(\tau)}$$

$$\mathcal{E}(\tau) = 4\pi i \partial_\tau \log [\eta(\tau) \sqrt{\tau_2}].$$

Using these definitions

$$\begin{aligned} \tilde{e}_1(\tau) &= \left[-4\pi i \partial_\tau \log \frac{\vartheta_s(0, \tau)}{\eta(\tau)} \right] - [4\pi i \partial_\tau \log [\eta(\tau) \sqrt{\tau_2}]] \\ &= -4\pi i \partial_\tau \left[\ln \vartheta_2(0|\tau) - \ln \eta(\tau) + \ln \eta(\tau) + \frac{1}{2} \ln \tau_2 \right] \end{aligned}$$

We have $\vartheta_2(0|\tau) = 2q^{\frac{1}{8}} \prod_{x \geq 1} (1 - q^x)(1 + q^x)^{2^1}$.

$$\begin{aligned} \tilde{e}_1(\tau) &= -4\pi i \partial_\tau \left[\ln 2 + \ln q^{\frac{1}{8}} + \sum_{x \geq 1} \ln(1 - q^x) + \sum_{x \geq 1} 2 \ln(1 + q^x) + \frac{1}{2} \ln \tau_2 \right] \\ \tilde{e}_1(\tau) &= \pi^2 - \frac{\pi}{\tau_2} + 8\pi^2 \sum_{x \geq 1} x q^x \left[-\frac{1}{1 - q^x} + \frac{2}{1 + q^x} \right] \\ A(\tau) &:= 8\pi^2 \sum_{x \geq 1} x q^x \left[-\frac{1}{1 - q^x} + \frac{2}{1 + q^x} \right] \quad B(\tau_2) := \pi^2 - \frac{\pi}{\tau_2}. \end{aligned}$$

Working with the holomorphic part we can use the geometric series

$$\begin{aligned} -\frac{1}{1 - q^x} + \frac{2}{1 + q^x} &= \sum_j -q^{xj} + 2(-q^x)^j \\ &= \sum_j q^{xj} (-1 + 2(-)^j). \end{aligned}$$

¹[**TODO**] Double-check the leading power of q in this product formula against the standard definition.

Thus, we have

$$A(\tau) = 8\pi^2 \sum_{x \geq 1} \sum_{j=0}^{\infty} x q^{(j+1)x} \left(2(-)^^j - 1 \right).$$

Which gives

$$\begin{aligned} \tilde{e}_1(\tau) &= 8\pi^2 \sum_{x \geq 1} \sum_{j=0}^{\infty} x q^{(j+1)x} \left(2(-)^^j - 1 \right) + \pi^2 - \frac{\pi}{\tau_2} \\ \tilde{e}_1(\tau) &= A(\tau) + B(\tau_2) \end{aligned}$$

In the integrand we have terms of the form $\tilde{e}_1^n(\tau)$ thus we have nested binomials

$$\begin{aligned} W(q) &:= \tilde{e}_1^n(\tau) = \sum_{p=0}^n \binom{n}{p} A(\tau)^p B^{n-p} \\ B^{n-p} &= \sum_{l=0}^{n-p} \binom{n-p}{l} (\pi^2)^{(n-p-l)} (-\pi)^l \tau_2^{-l}. \end{aligned}$$

The holomorphic part of the Weierstrass contribution is

$$W(q) = \sum_{p=0}^n \sum_{l=0}^{n-p} \binom{n}{p} \binom{n-p}{l} \left(\sum_{x \geq 1} \sum_{j=0}^{\infty} 8\pi^2 x q^{(j+1)x} \left(2(-)^^j - 1 \right) \right)^p \left((\pi^2)^{n-p-l} (-\pi)^l \tau_2^{-l} \right) \quad (4.37)$$

We can define

$$S(\tau) := 8\pi^2 \sum_{m \geq 1} c(m) q^m, \quad c(m) = \sum_{\substack{x \geq 1, j \geq 0 \\ (j+1)x = m}} x \left(2(-1)^j - 1 \right) \quad (4.38)$$

$$S^p = (8\pi^2)^p \sum_{M \geq 0} \sum_{\substack{m_1 + \dots + m_p = M \\ m_i \geq 1}} \prod_{\varpi=1}^p c(m_{\varpi}) q^M \quad (4.39)$$

Define $C_p(M) := \sum_{\substack{m_1 + \dots + m_p = M \\ m_i \geq 1}} \prod_{\varpi=1}^p c(m_{\varpi})$. Then we have

$$S^p = (8\pi^2)^p \sum_{M \geq 0} C_p(M) q^M \quad (4.40)$$

Thus, explicitly we have the holomorphic contribution

$$W(q) = \sum_{p=0}^n \sum_{l=0}^{n-p} \binom{n}{p} \binom{n-p}{l} (8\pi^2)^p \sum_{M \geq 0} C_p(M) q^M \left(\pi^{2((n-p)-l)} (-\pi)^l \tau_2^{-l} \right).$$

Then, if we consider the conjugate, we get a net contribution of

$$W(q, \bar{q}) = \sum_{p=0}^n \sum_{l=0}^{n-p} \sum_{\bar{p}=0}^{\bar{n}} \sum_{\bar{l}=0}^{\bar{n}-\bar{p}} \sum_{M, \bar{M} \geq 0} (8\pi^2)^{p+\bar{p}} \binom{n}{p} \binom{(n-p)}{l} \binom{\bar{n}}{\bar{p}} \binom{(\bar{n}-\bar{p})}{\bar{l}} C_p(M) \bar{C}_{\bar{p}}(\bar{M}) \\ \left(\pi^{2((n-p)-l)} (-\pi)^l \tau_2^{-l} \right) \left(\pi^{2((\bar{n}-\bar{p})-\bar{l})} (-\pi)^{\bar{l}} \tau_2^{-\bar{l}} \right) q^M \bar{q}^{\bar{M}}.$$

4.4.2 Alternative Weierstrass Contribution

The previously defined Weierstrass contribution has a series without a constant term, making it unwieldy for systematic numerical manipulation. Here we redefine the series so that it incorporates a constant term.

Taking the previous definition,

$$\tilde{e}_1(\tau) = \pi^2 - \frac{\pi}{\tau_2} + 8\pi^2 \sum_{x \geq 1} x q^x \left[-\frac{1}{1-q^x} + \frac{2}{1+q^x} \right] \\ A(\tau) := \pi^2 + 8\pi^2 \sum_{x \geq 1} x q^x \left[-\frac{1}{1-q^x} + \frac{2}{1+q^x} \right] \quad B(\tau_2) := -\frac{\pi}{\tau_2}.$$

Raising to the n -th power leaves the following holomorphic part

$$W(q) = \sum_{p=0}^n \sum \binom{n}{p} \left(\pi^2 + \sum_{x \geq 1} \sum_{j \geq 0} 8\pi^2 x q^{(j+1)x} \left(2(-)^j - 1 \right) \right)^{n-p} \left((-\pi)^p \tau_2^{-p} \right) \quad (4.41)$$

4.4.3 The vertex normalisation contribution $V(q, \bar{q})$.

One may use the identity

$$\vartheta'_1(0|\tau) = 2\pi\eta^3(\tau) \quad \eta(\tau) := q^{\frac{1}{24}} \prod_{i=1}^{\infty} (1 - q^n) \quad (4.42)$$

So,

$$\vartheta'_1(0|\tau) = 2\pi q^{\frac{1}{8}} \prod_{i=1}^{\infty} (1 - q^n)^3 \\ |\vartheta'_1(0|\tau)|^{4(N-1)} = (2\pi)^{4(N-1)} q^{\frac{N-1}{4}} \bar{q}^{\frac{N-1}{4}} \prod_{i=1}^{\infty} (1 - q^n)^{6(N-1)} \prod_{i=1}^{\infty} (1 - \bar{q}^n)^{6(N-1)}.$$

If we define $\hat{\eta}(\tau) := \prod_{i=1}^{\infty} (1 - q^n)^{6(N-1)}$, then we have

$$|\vartheta'_1(0|\tau)|^{4(N-1)} = (2\pi)^{4(N-1)} q^{\frac{N-1}{4}} \bar{q}^{\frac{N-1}{4}} \hat{\eta}(\tau) \hat{\bar{\eta}}(\tau) \quad (4.43)$$

The contribution to the amplitude takes the form of the inverse of 4.43. We can expand $\frac{1}{\hat{\eta}} = \sum_{j \geq 0} \tilde{c}_j q^j$. Thus, we have

$$\frac{1}{|\vartheta'_1(0|\tau)|^{4(N-1)}} = (2\pi)^{-4(N-1)} q^{-\frac{N-1}{4}} \bar{q}^{-\frac{N-1}{4}} \sum_{j \geq 0} \tilde{c}_j q^j \sum_{\bar{j} \geq 0} \tilde{\bar{c}}_{\bar{j}} \bar{q}^{\bar{j}}.$$

If we denote the vertex normalisation contribution (excluding the factor of 2π) by $V(q, \bar{q})$ ² we may only consider the holomorphic part

$$V(q) = q^{-\frac{N-1}{4}} \sum_{j \geq 0} \tilde{c}_j q^j \quad (4.44)$$

We want to find a condition on the $\tilde{c}_j$. If we impose that ³

$$\begin{aligned} \hat{\eta}(\tau) \cdot \frac{1}{\hat{\eta}(\tau)} &= 1 \quad \implies \left[\sum_{r \geq 0} c_r^{\hat{\eta}} q^r \right] \left[\sum_{j \geq 0} \tilde{c}_j q^j \right] = 1 \\ \left[\sum_{r \geq 0} c_r^{\hat{\eta}} q^r \right] \left[\sum_{j \geq 0} \tilde{c}_j q^j \right] &= \sum_{n \geq 0} \sum_{r=0}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} q^n \\ \sum_{r=0}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} &= \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \geq 1 \end{cases} \\ &\implies c_0^{\hat{\eta}} \tilde{c}_0 = 1. \end{aligned}$$

However, we know $c_0^{\hat{\eta}}$ is the constant term of $\hat{\eta}$ which by definition is 1. Thus, $\tilde{c}_0 = 1$. $\forall n \geq 1$

$$\begin{aligned} \sum_{r=0}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} &= 0 \\ c_0^{\hat{\eta}} \tilde{c}_n + \sum_{r=1}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} &= 0 \\ \tilde{c}_n &= - \sum_{r=1}^n c_r^{\hat{\eta}} \tilde{c}_{n-r} \quad \tilde{c}_0 = 1. \end{aligned}$$

So we have the holomorphic contribution

$$V(q) := q^{-\frac{N-1}{4}} \sum_{j \geq 0} \tilde{c}_j q^j \quad \text{with } \tilde{c}_0 = 1, \quad \tilde{c}_j = - \sum_{r=1}^j c_r^{\hat{\eta}} \tilde{c}_{j-r}.$$

and the net vertex normalisation contribution

$$V(q, \bar{q}) := \frac{1}{|\vartheta'_1(0|\tau)|^{4(N-1)}} = (2\pi)^{-4(N-1)} q^{-\frac{N-1}{4}} \bar{q}^{-\frac{N-1}{4}} \sum_{j, \bar{j} \geq 0} \tilde{c}_j \tilde{\bar{c}}_{\bar{j}} q^j \bar{q}^{\bar{j}}.$$

²**[TODO]** Revisit whether the 2π factor should be folded back into V before it's used elsewhere.

³**[TODO]** Confirm the sign of the power here (i.e. that this is $\hat{\eta}^{-1}$, not $\hat{\eta}^{+1}$) is the right normalisation condition.

4.4.4 The f_1 Theta contribution

We have

$$f_1(\tau) := \left(\frac{\vartheta'_1(0|\tau)}{\vartheta_2(0|\tau)} \right)^2 = 8\pi^2 \left(\frac{1}{8} - q + 3q^2 - 4q^3 + \dots \right) \quad (4.45)$$

⁴. Thus we have

$$f_1(\tau) = \sum_{r \geq 0} b_r q^r \quad (4.46)$$

and in raising to a power we use the Cauchy product

$$f_1(\tau)^{k-n} = \left(\sum_{r \geq 0} b_r q^r \right)^{k-n} = \sum_{i \geq 0} \left[\sum_{r_1+r_2+\dots+r_{k-n}=i} b_{r_1} b_{r_2} \cdot b_{r_{k-n}} \right] q^i \quad (4.47)$$

Define $d_i^{k-n} := \sum_{r_1+r_2+\dots+r_{k-n}=i} b_{r_1} b_{r_2} \cdot b_{r_{k-n}}$. Thus,

$$f_1(\tau)^{k-n} = \sum_{i \geq 0} d_i^{k-n} q^i \quad (4.48)$$

4.4.5 The Lattice contribution $L(q, \bar{q})$

We have

$$\mathcal{A}^{(N)} \sim \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^{\frac{9}{2}}} |\vartheta'_1(0|\tau)|^{4(N-1)} \sum_k \dots \mathcal{I}_{2n+2, \dots, \dots}(\tau, \bar{\tau}) \quad (4.49)$$

with

$$\begin{aligned} \mathcal{I}_{N_1, \bar{N}_1, N_2, \bar{N}_2}(\tau, \bar{\tau}) &:= \frac{(-)^{\frac{N_1+\bar{N}_1}{2}}}{2\sqrt{\tau_2(N-1)}} \sum_{h=1}^{2(N-1)} \sum_{\mathbf{K} \in \mathbb{Z}^{2(N-1)}} \sum_{\bar{\mathbf{K}} \in \mathbb{Z}^{2(N-1)}} \delta_{\mathbf{1}_{2(N-1)} \cdot \mathbf{K}, h} \delta_{\mathbf{1}_{2(N-1)} \cdot \bar{\mathbf{K}}, h} \\ &\quad q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4(N-1)}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} q^{\frac{1}{2} \bar{\mathbf{K}} \cdot \bar{\mathbf{K}} - \frac{h^2}{4(N-1)}} e^{-i\pi \bar{\mathbf{a}} \cdot \bar{\mathbf{K}}}. \end{aligned}$$

$$\mathcal{I}_{2n+2, \dots, \dots}(\tau, \bar{\tau}) \sim \tau_2^{-\frac{1}{2}} \cdot \text{lattice} \quad (4.50)$$

If we combine the factor of $\tau_2^{-\frac{1}{2}}$ into the $\tau_2^{\frac{9}{2}}$ we have a holomorphic part remaining. Thus, define,

$$\theta_h^{[N_1]}(q) := \sum_{\substack{\mathbf{K} \in \mathbb{Z}^{2(N-1)}, \\ \mathbf{1} \cdot \mathbf{K} = h}} q^{\frac{1}{2} \mathbf{K} \cdot \mathbf{K} - \frac{h^2}{4(N-1)}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \quad (4.51)$$

Fixing N_1 fixes a and h .

$$g_h^{[N_1]} = \sum_{\substack{\mathbf{K} \in \mathbb{Z}^{2(N-1)} \\ \mathbf{1} \cdot \mathbf{K} = h \\ \mathbf{K}^2 = 2m_{Lat} + h^2/(2N-2)}} e^{i\pi \mathbf{a} \cdot \mathbf{K}} \quad (4.52)$$

⁴A.25 Bianchi

Define $m_{\text{lat}} := \frac{1}{2}k \cdot k - \frac{h^2}{4(N-1)}$

$$\theta_h^{[N_1]}(q) = \sum_{m_{\text{lat}}} g_h^{N_1}(m_{\text{lat}}) q^{m_{\text{lat}}} \quad (4.53)$$

So the lattice part can be written as

$$\begin{aligned} L(q, \bar{q}) &= \frac{(-1)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{(N-1)}} \sum_{h=1}^{2(N-1)} \theta_h^{[N_1]}(q) \bar{\theta}_h^{[N_1]}(\bar{q}) \\ L(q, \bar{q}) &= \frac{(-1)^{\frac{N_1 + \bar{N}_1}{2}}}{2\sqrt{(N-1)}} \sum_{h=1}^{2(N-1)} \sum_{m_{\text{lat}}, \bar{m}_{\text{lat}}} g_h^{N_1}(m_{\text{lat}}) \bar{g}_h^{\bar{N}_1}(\bar{m}_{\text{lat}}) q^{m_{\text{lat}}} \bar{q}^{\bar{m}_{\text{lat}}}. \end{aligned}$$

where $N_1 = 2n + 2$ and $\bar{N}_1 = 2\bar{n} + 2$.

4.4.6 Collecting the contributions

$$\begin{aligned} \mathcal{A}^{(N)} &= \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^{\frac{9}{2}} |\vartheta_1'(0|\tau)|^{4(N-1)}} \sum_{k=0}^{N-2} \binom{N-2}{k}^2 \left(\frac{\pi}{\tau_2}\right)^{2(N-2-k)} \sum_{n, \bar{n}=0}^k \binom{k}{n} \binom{k}{\bar{n}} \\ &\quad \tilde{e}_1(\tau)^n \bar{\tilde{e}}_1(\tau)^{\bar{n}} f_1(\tau)^{k-n} \bar{f}_1(\tau)^{k-\bar{n}} \mathcal{I}_{2n+2, 2\bar{n}+2, 2N-4, 2N-4-2\bar{n}}(\tau, \bar{\tau}). \end{aligned}$$

Inspecting the powers of q we can define $m := -\frac{N-1}{4} + j + M + i + m_{\text{lat}}$. Likewise, for τ_2 , define $\ell = l + \bar{l}$.

$$F_\ell(m, \bar{m}) := \sum_{\substack{k, n, \bar{n} \\ p, \bar{p}, l + \bar{l} = \ell}} \binom{N-2}{k}^2 \binom{k}{n} \binom{k}{\bar{n}} \binom{n}{p} \binom{n-p}{l} \binom{\bar{n}}{\bar{p}} \binom{\bar{n}-\bar{p}}{\bar{l}} (8\pi^2)^{p+\bar{p}} \pi^{2(n-p-l)+2(\bar{n}-\bar{p}-\bar{l})} (-\pi)^\ell \quad (4.54)$$

$$\frac{(-1)^{n+\bar{n}}}{2\sqrt{N-1}} \sum_{h=1}^{2(N-1)} \mathcal{B}_{k, n, p, l}^{(h)}(m) \mathcal{B}_{k, \bar{n}, \bar{p}, \bar{l}}^{(h)}(\bar{m}) \quad (4.55)$$

where

$$\mathcal{B}_{k, n, p, l}^{(h)}(m) = \sum_{\substack{j, M, i, m_{\text{lat}} \geq 0 \\ -\frac{N-1}{4}, j+M+i+m_{\text{lat}}=m}} \tilde{c}_j C_p(M) d_i^{(k-n)} g_h^{2n+2}(m_{\text{lat}}).$$

Then we have

$$\mathcal{A}^{(N)} = (2\pi)^{-4(N-1)} \int_{\mathcal{F}} d^2\tau \tau_2^{-\frac{9}{2}} \sum_{\ell=0}^{2(N-1)} \tau_2^{-\ell} \sum_{m, \bar{m}} F_\ell(m, \bar{m}) q^m \bar{q}^{\bar{m}}.$$

5 An Analytical Description of the Decay Width

5.1 Identifying the contributions

In looking at the imaginary part, we notice it is equivalent to the negative powers of q which is exactly the values which make m negative, where

$$m = -\frac{N-1}{4} + j + m_{\text{lat}} + i + M \quad (5.1)$$

and equivalently for the anti-holomorphic part.

Since $m \geq 0$, we know $j \in \{0, \frac{N-1}{4}\}$.¹ To find a bound on m :

$$\forall k_a^2 \in \mathbb{Z},$$

$$k_a^2 = k_a \pmod{2} \quad (5.2)$$

$$\sum_a k_a^2 - h \in 2\mathbb{Z} \quad (5.3)$$

$$\sum_a k_a^2 = \sum_a [k_a + k_a(k_a - 1)] \quad (5.4)$$

let $s \in \mathbb{Z}_{\geq 0}$, then set $\sum_a k_a(k_a - 1) = 2s$.

$$\begin{aligned} m_{\text{lat}} &= \frac{1}{2} \vec{k}^2 - \frac{h^2}{4(N-1)} \\ &= \frac{1}{2} [h + 2s] - \frac{h^2}{4(N-1)} \\ &= \frac{[2(N-1) - h]h}{4(N-1)} + s. \end{aligned}$$

So we have

$$\begin{aligned} j + i + M + m_{\text{lat}} &< \frac{N-1}{4} \\ j + i + M + \frac{[2(N-1) - h]h}{4(N-1)} + s &< \frac{N-1}{4} \\ j + i + M + s &< \frac{(N-1)^2 - 2(N-1)h + h^2}{4(N-1)} \\ j + i + M + s &< \frac{[h - (N-1)]^2}{4(N-1)}. \end{aligned}$$

So if we define $K := j + i + M + s$ we have

$$m = K - \frac{[h - (N-1)]^2}{4(N-1)} \quad (5.5)$$

¹**[TODO]** Double check this range for j as written it restricts j to two isolated points rather than an interval, which looks off; it should probably read $0 \leq j \leq \frac{N-1}{4}$.

Thus, we have negative q contributions for

$$K \in \left[0, 1, \dots, \frac{[h - (N - 1)]^2}{4(N - 1)} - 1 \right] \quad (5.6)$$

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